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# First paleomagnetic and $^{40}Ar/^{39}Ar$ study of Paleoproterozoic rocks from the French Guyana (Camopi and Oyapok rivers), northeastern Guyana Shield

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**Abstract**

In order to understand the Paleoproterozoic geographic evolution of the Guyana Shield, paleomagnetic and $^{40}Ar/^{39}Ar$ investigations were carried out on granitoids and volcano-sedimentary rocks from the Oyapok and Camopi rivers (French Guyana–Brazil frontier). Scanning electronic microscope, thermomagnetic and isothermomagnetic experiments show that magnetite is the main magnetic remanent carrier in most of the samples. The metavolcano-sedimentary rocks (Paramaca) show a weak magnetization and scattered magnetic directions. Therefore, no reliable magnetic component could be isolated from these samples. Samples taken from tonalite and meta-ultrabasite rocks yield a characteristic magnetic direction, carried by subautomorphous magnetite, that is well defined and distinct from that of the present Earth field and that of nearby Jurassic dikes. A virtual geomagnetic pole (VGP) deduced from this probable primary remanence was calculated, namely pole OYA, $\lambda = 28.0°S$, $\phi = 346.0°E$, $N = 5$, $k = 31.9$ and $A_{95} = 13.8°$. Four $^{40}Ar/^{39}Ar$ ages, ranging from 2052 to 1973 Ma, were obtained from amphiboles and biotites of tonalite rocks, showing a relatively slow cooling rate of ca $4.8 + 2.6/-2.1°C\ Ma^{-1}$. The linear extrapolation of this cooling rate to the magnetite unblocking temperature (540 to 580°C) yields a magnetization age of $2036 \pm 14$ Ma for pole OYA. Pole OYA differs significantly from available paleomagnetic results from Venezuela of the West Guyana Shield dated at $2000 \pm 10$ Ma. This difference may indicate an important latitudinal movement of the Guyana Shield between 2036 and 2000 Ma with a velocity of $9 \pm 7$ cm/year. © 2001 Elsevier Science B.V. All rights reserved.

*Keywords:* Guyana Shield; paleomagnetism; $^{40}Ar/^{39}Ar$ dating; Paleoproterozoic

## 1. Introduction

Paleomagnetism, for decades, has provided important constraints on the geodynamic history of the Earth. Most of the available data (77%) were obtained on Phanerozoic rocks (Global Paleo-

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magnetic Database version 3.3; GPMDB, 1998). Only 7% concern the Paleoproterozoic period, which occupies 20% of Earth history. Recent paleomagnetic data concerning the Fennoscandian Shield (Torsvik and Meert, 1995; Fedotova et al., 1999; Mertanen et al., 1999) and North American Shield (Buchan et al., 1996) yielded new paleogeographic constraints between these two zones during this period. Most of these investigations were conducted on intrusive rocks and were associated with U/Pb dating (Buchan et al., 1996; Fedotova et al., 1999; Mertanen et al., 1999). The magnetic remanence age in quickly cooled volcanic rocks and dikes is close to the U/Pb age, but in slowly cooled Precambrian terrains the age of acquisition of thermal remanent magnetization (TRM) cannot always be directly defined by one isotopic method. The slow cooling rate is generally attributed to gradual uplift and erosion. In order to estimate the magnetic remanence age more precisely, the combined study of the paleomagnetism with $^{40}Ar/^{39}Ar$ thermochronology is important and sometimes critical (Berger, 1979; Costanzo-Alvarez and Dunlop, 1988; Briden et al., 1993).

In the framework of a multidisciplinary BRGM (Bureau de Recherche Géologique et Minière) geological mapping project of the French Guyana Territory in collaboration with Orleans Institut of Geosciences (ISTO), UMR-CNRS 6527 and the CPRM (Brazilian geological survey), we carried out two field trips in 1997 and 1998. This study presents new time-calibrated paleomagnetic data from the northeastern part of the Guyana Shield. The $^{40}Ar/^{39}Ar$ data allow one to evaluate the cooling rate for the central Oyapok/Camopi zone at about 2 Ga for the first time. Paleogeographic and tectonic implications of paleomagnetic results are then discussed with previous studies from the northwest part of the Guyana Shield (Fig. 1, Onstott and Hargraves, 1981; Onstott et al., 1984, 1989)

## 2. Geological setting and sampling

### 2.1. Regional and local geology

The Guyana Shield is composed of a narrow Archean belt (Imataca Complex in Venezuela) and of granite–greenstone belts (2.2 to 2.0 Ga), formed during the Transamazonian tectonothermal event (Fig. 1; Montgomery and Hurley, 1978; Teixieira et al., 1989). Two major strike-slip fault zones cross the Guyana Shield: (1) the Guri fault zone (GFZ) separates the Archean belt from the Paleoproterozoic series; (2) the Pisco Jurua fault (PJF) is located on the western border of the Central Guyana Granulite Belt (CGGB) and extends from western Suriname to Roraima State, Brazil (Fig. 1), thus dividing the shield into two parts (Fig. 1). The PJF is interpreted as a mobile belt reflected by the charnockitization of older rocks (Dahlberg, 1987; Gibbs and Baron, 1993). French Guyana is situated in the northeastern part of the Guyana Shield (Fig. 1). The Paleoproterozoic lithological succession is well known in the Oyapok–Camopi rivers; the basement is composed of the Paramaca greenstone belt sequence (Fig. 2). It is constituted by volcanic rocks with tholeiitic and calc-alkaline compositions (Milési et al., 1995; Egal et al., 1995) dated at $2110 \pm 90$ Ma (Sm/Nd age, Gruau et al., 1985). These volcanic rocks are covered by sedimentary deposits of Armina (Fig. 2, Ledru et al., 1991). The volcano-sedimentary rocks are metamorphosed and deformed by the Guyana and Caribbean granite intrusions (Fig. 2, Choubert, 1974), which are dated at 2130 to 2080 Ma (Pb/Pb ages, Vanderhaeghe et al., 1998). All these rocks are crosscut by Mesozoic dikes (Fig. 2) associated with the Central Atlantic Ocean prerifting stages (Deckart et al., 1997). This latest volcanic event constitutes the only major thermal event since the Early Paleoproterozoic in French Guyana.

### 2.2. Paleomagnetic and geochronological sampling

Paleomagnetic samples were collected in 16 sites along the Oyapok and Camopi rivers (Fig. 2 and Table 1) from tonalite (five sites), amphibolites and metasediments (ten sites) and from metaultrabasites (one site). The tonalite has a coarse-grained texture and is composed of magnesiohornblende, biotite, plagioclase ($An_{30}$–$An_{60}$), interstitial quartz and titanite as the principal accessory minerals. In each site, six or seven cores

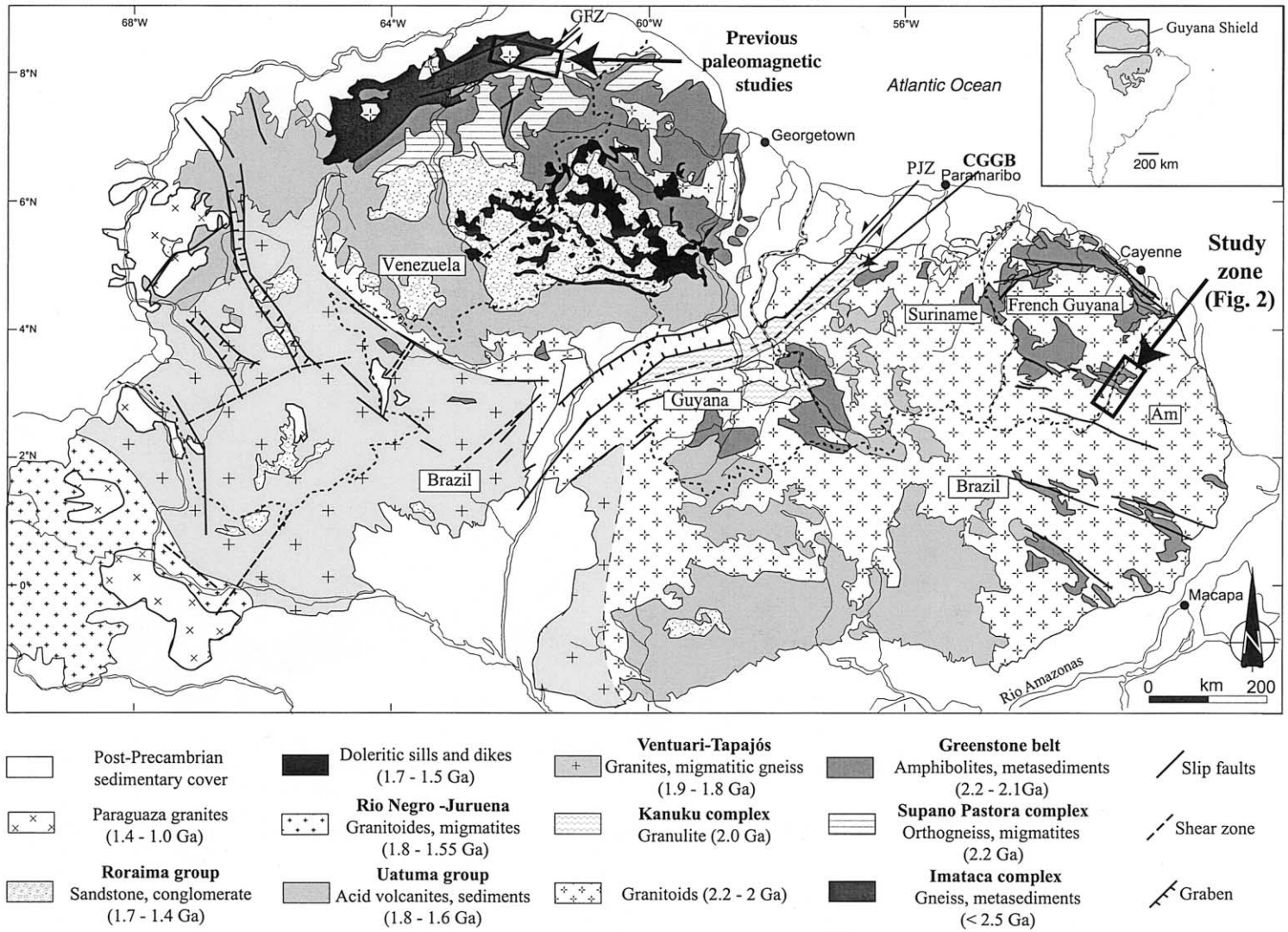



Fig. 1. Geological map of Guyana Shield (modified after Gibbs and Baron (1993)). CGGB: Central Guyana Granulitic belt. GFZ: Guri fault zone; PJZ: Pisco Jurua fault zone. Our study zone and previous paleomagnetic sampling are indicated.

Table 1
Summary of paleomagnetic sampling and analysis results. Slat. and Slong.: latitude and longitude of the site; Dec., Inc., $k$, $\alpha_{95}$ ($A_{95}$): declination, inclination, precision parameter and confidence interval at the 95% level; $n/N$: number of entries in statistics/number of treated specimens; Plat. and Plong.: latitude and longitude of polar coordinates[a]

| Site | Rock | Slat. | Slong. | $n/N$ | Dec. | Inc. | Plat. | Plong. | $k$[a] | $\alpha_{95}$ | $A_{95}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| *Paramaca rocks* | | | | | | | | | | | |
| GC | amphibolite | 2° 59′ 42″ | −52° 21′ 58″ | | | | | | scattered | | |
| GE | metasediment | 3° 10′ 33″ | −52° 29′ 10″ | 4/7 | 15.9° | 26.7° | | | 17.3 | 15.3° | |
| GF | metasediment | 3° 12′ 43″ | −52° 27′ 22″ | 3/6 | 339.6° | 19.9° | | | 10.3 | 30.0° | |
| GG | amphibolite | 3° 13′ 00″ | −52° 23′ 49″ | | | | | | scattered | | |
| GI | amphibolite | 3° 12′ 13″ | −52° 24′ 24″ | | | | | | scattered | | |
| GJ | metasediment | 3° 12′ 59″ | −52° 23′ 50″ | | | | | | scattered | | |
| GK | metasediment | 3° 10′ 45″ | −52° 20′ 02″ | 4/6 | 348.3° | 38.7° | | | 13.5 | 26.0° | |
| GL | amphibolite | 3° 10′ 35″ | −52° 21′ 55″ | | | | | | scattered | | |
| GM | amphibolite | 3° 07′ 17″ | −52° 20′ 54″ | 4/7 | 348.1° | 26.2° | | | 16.5 | 15.3° | |
| GP | amphibolite | 3° 13′ 13″ | −52° 17′ 37″ | 4/6 | 339.1° | 3.4° | | | 10.5 | 29.8° | |
| Means | | | | 5/10 | 348.9° | 26.6° | | | 18.7 | 18.2° | |
| *Meta-ultrabasite and tonalite* | | | | | | | | | | | |
| GH | Meta-ultrabasite | 3° 12′ 34″ | −52° 26′ 38″ | 6/6 | 157.9° | 54.4° | −46° | 334.5° | 177.3 | 4.1° | |
| GN | tonalite | 3° 08′ 07″ | −52° 20′ 47″ | 7/8 | 136.0° | 56.6° | −33° | 349.9° | 33.5 | 9.5° | |
| PQ | tonalite | 3° 08′ 10″ | −52° 20′ 41″ | 6/6 | 120.2° | 63.0° | −19° | 348.8° | 28.9 | 9.7° | |
| PR | tonalite | 3° 08′ 27″ | −52° 20′ 36″ | 5/6 | 137.2° | 67.6° | −26° | 336.8° | 58.8 | 7.9° | |
| PS[b] | tonalite | 3° 08′ 34″ | −52° 20′ 27″ | 6/6 | 330.0° | 75.2° | −27° | 292.6° | 70.2 | 5.0° | |
| PT | tonalite | 3° 08′ 40″ | −52° 20′ 14″ | 6/7 | 116.2° | 54.7° | −19° | 358.7° | 60.5 | 7.8° | |
| Means OYA | | | | 5/6 | 133.8° | 60.2° | | | 59.9 | 9.9° | |
| | | | | | | | −28.0° | 346.0° | | | 13.8° |

[a] 'Scattered' indicates a non-Fisherian distribution.
[b] Site excluded from mean calculation.

were drilled. Core orientation was measured by magnetic compass and, whenever possible, by sun orientation. The average difference between solar and magnetic azimuth measurements is about $18 \pm 2°$, which is compatible with the 1998 IGRF declination (17.5°) for this location. Cores were cut to standard size (2.54/2.2 cm), yielding about 100 specimens. Two sites were sampled for the geochronological investigation. One corresponds to the paleomagnetic site GN and, in order to constrain the cooling rate, another (B107) was sampled 20 km farther north in a separate pluton (Fig. 2).

## 3. Laboratory measurements and analytical procedures

### *3.1. Magnetic mineralogical analysis and petrographic study*

To characterize magnetic mineral compositions,

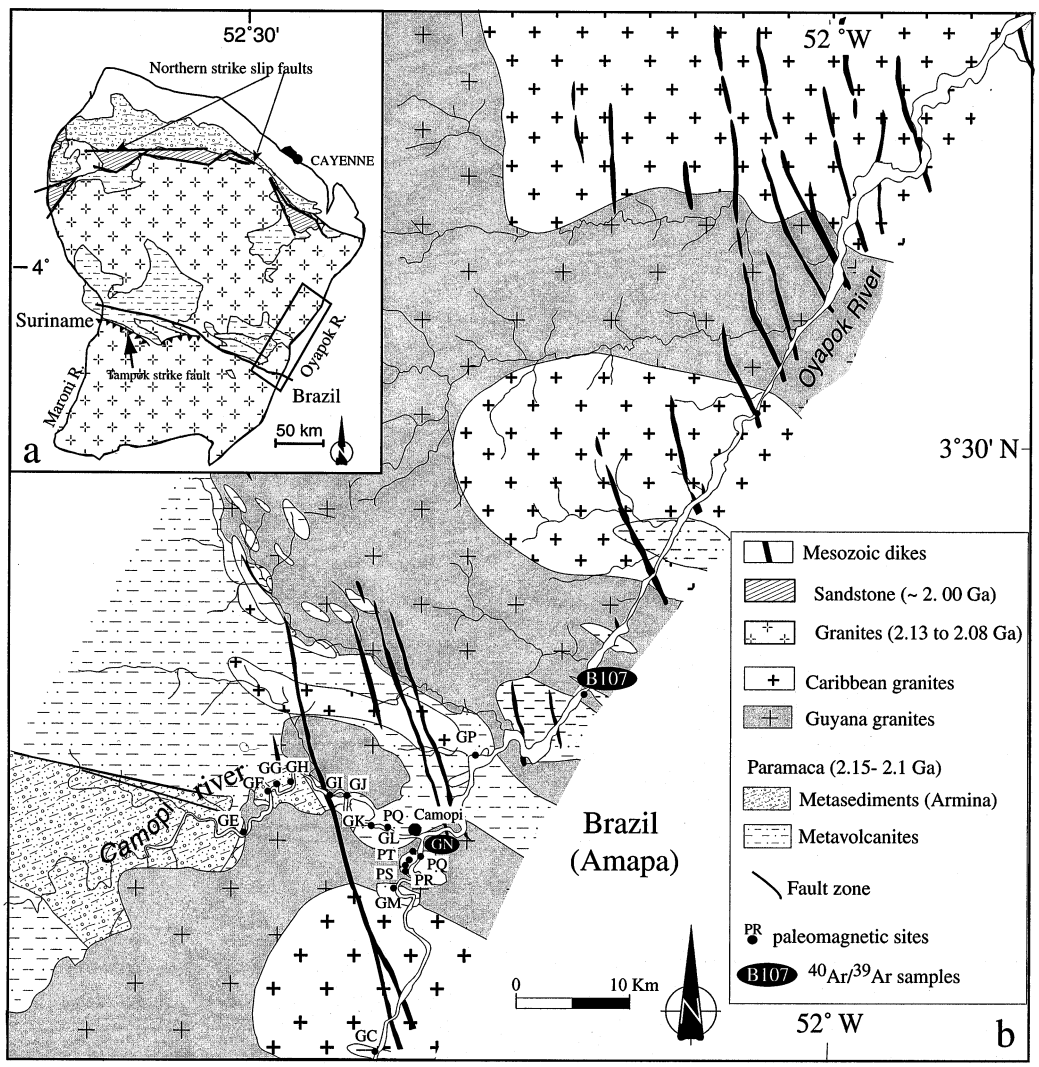



Fig. 2. (a) Simplified geological map of French Guyana (modified after Ledru et al. (1991)); (b) Geological map of the Oyapok and Camopi rivers zone (modified after Marot (1988)). Sampling location for paleomagnetic and geochronological studies are indicated.

### 3.1. Magnetic mineralogical analysis and petrographic study

To characterize magnetic mineral compositions, we applied the following methods on representative samples: reflection microscopy (Olympus BX60) at the geological laboratory of Université d'Orléans; scanning electronic microscopy (SEM, JEOL) at Ecole Supérieur de l'Energie et des Matériaux (ESEM) in Orléans; thermomagnetic experiments using a CS3 apparatus coupled with a KLY-3S kappabridge (AGICO, Geofysica) at the joint BRGM/Université d'Orléans Laboratoire de Magnétisme des Roches (LMR). Isothermal remanent magnetization was measured with an IM-10 impulse magnetizer and a JR5 spinner magnetometer (AGICO, Geofysica), and the latter was also used for remanent magnetization measurements. Hysteresis loops were realized with a translation inductiometer within an electromagnet providing a field of up to 1.5 T at the Paleomagnetic Laboratory of Saint Maur (Paris).

### 3.2. $^{40}Ar/^{39}Ar$ analytical procedures

$^{40}Ar/^{39}Ar$ ages were measured at the geochronological laboratory of Nice. Single grains of biotite and amphibole were separated by heavy liquids and finally isolated by hand picking (in BRGM). They were carefully selected under a binocular microscope from the coarser fractions. Grain sizes varied from 200 to 400 μm. The samples were irradiated at McMaster Nuclear Reactor in the University of Hamilton, Canada, with total flux of $3 \times 10^{18}$ n cm$^{-2}$. The maximum flux gradient is estimated at $\pm 0.2\%$ in the volume where the samples were included. The irradiation standard was the Hb3gr hornblende as a flux monitor with age of 1072 Ma (Turner et al., 1971; recently confirmed by Renne, 2000). The classical step-heating procedure is described by Féraud et al. (1982, 1986) and performed with a laser probe using a Coherent Innova 70-4 continuous argon-ion laser model. The mineral was located on a copper sample-holder, beneath a Pyrex window. Each step-heating laser experiment lasted about 4 min: 1 min of laser heating, 2 min of clean-up in the purification line, consisting of a getter operating at 400°C and an $N_2$ cold trap, and finally 40 s of inlet time into the mass spectrometer. The temperature is not known, but its homogeneity was controlled by binocular microscope coupled with a color video camera and recorder. The mass spectrometer was a VG3600 working with a Daly detector system. Isotopic measurements were corrected for K, Ca and Cl isotopic interferences, mass discrimination, and atmospheric argon contamination. The measured $^{40}Ar/^{39}Ar$ atmospheric ratio, determining the mass discrimination, was $289.3 \pm 1$ during the analyses.

To define a plateau age, at least three consecutive steps are needed, corresponding to a minimum of 70% of total $^{39}Ar_K$ released, and the individual fraction ages should agree within $2\sigma$ with the 'integrated' age of the plateau segment. All uncertainties are quoted at the $2\sigma$ (except the apparent ages in Table 2 and age spectra, which are given at the $1\sigma$ level) and do not include the uncertainties on the age of the monitor. The uncertainties on the $^{40}Ar^*/^{39}Ar_K$ ratios of the monitor are included in the calculation of the plateau age uncertainty.

## 4. Magnetic mineralogy

Petrographic observations in transmitted light show a considerable amount of subautomorphous grains of magnetite (Fig. 3a), with xenomorphous grains of ilmenite, pyrite in tonalite and meta-ultrabasite. However, in Paramaca rocks, elongate ilmenite consists of the principal ferri-oxide and no magnetite grains were observed (Fig. 3b). The saturation field for isothermal remanent magnetization (IRM) is 0.3 T (Fig. 4a) for tonalitic rock. The magnetic saturation intensity for tonalite samples is greater than 30 A m$^{-1}$ (Fig. 4a). Thermomagnetic curves show sharp drops for tonalite and meta-ultrabasite rocks at around 560–580°C (samples GN4 and GH1, respectively; Fig. 4b), and do not show a drop for the Paramaca (samples GM4; Fig. 4c). Hysteresis loops performed on the tonalite rocks showed narrow-waisted and typical low coercivity magnetite grains (Fig. 4d; Raposo and D'Agrella-Filho, 2000). In contrast, the Paramaca samples presented hysteresis curves

Table 2
Detailed $^{40}Ar/^{39}Ar$ analytical results. $^{40}Ar^*$: radiogenic $^{40}Ar$; Ca and K: produced by Ca and K neutron interference, respectively. Decay constants are those of Steiger and Jäger (1977). Correction factors for interfering isotopes were $(^{39}Ar/^{37}Ar)_{Ca} = 7.06 \times 10^{-4}$, $(^{36}Ar/^{37}Ar)_{Ca} = 2.79 \times 10^{-4}$, $(^{40}Ar/^{39}Ar)_K = 2.97 \times 10^{-2}$. Data given in italics are excluded from the age calculation for GN

| Laser step (mV) | Atmospheric (%) | $^{39}Ar_K$ (%) | $^{37}Ar_{Ca}/^{39}Ar_K$ (%) | $^{40}Ar^*/^{39}Ar_K$ (%) | Age (Ma) |
|---|---|---|---|---|---|
| **B107 amphibole** | | | | | |
| 59 | 12.10 | 0.04 | 14.89 | 645.63 | 5768.14 ± 358.96 |
| 157 | 1.03 | 2.46 | 3.69 | 66.17 | 2210.43 ± 10.69 |
| 169 | 0.00 | 3.53 | 3.67 | 59.11 | 2069.15 ± 10.39 |
| 177 | 1.58 | 6.39 | 3.72 | 57.33 | 2031.59 ± 32.45 |
| 189 | 0.01 | 11.47 | 3.74 | 58.42 | 2054.71 ± 3.78 |
| 198 | 0.01 | 39.27 | 3.72 | 58.36 | 2053.47 ± 2.73 |
| 208 | 0.22 | 5.41 | 3.70 | 58.33 | 2052.67 ± 7.95 |
| 228 | 0.00 | 6.30 | 3.70 | 58.47 | 2055.68 ± 7.68 |
| 287 | 0.55 | 5.36 | 3.72 | 57.67 | 2038.76 ± 8.42 |
| Fuse | 0.13 | 19.78 | 3.70 | 58.17 | 2049.39 ± 3.96 |
| Total age: | | | | | 2065.39 ± 4.40 |
| **B107 biotite** | | | | | |
| 18 | 61.90 | 1.25 | 0.09 | 19.99 | 985.48 ± 28.27 |
| 30 | 1.38 | 6.07 | 0.00 | 50.75 | 1885.96 ± 4.54 |
| 35 | 0.15 | 12.36 | 0.01 | 54.28 | 1965.56 ± 2.69 |
| 38 | 0.00 | 13.56 | 0.01 | 55.65 | 1995.59 ± 3.12 |
| 41 | 0.03 | 5.63 | 0.01 | 55.68 | 1996.16 ± 3.67 |
| 48 | 0.00 | 4.82 | 0.01 | 56.09 | 2005.05 ± 4.71 |
| 55 | 0.00 | 6.56 | 0.01 | 56.00 | 2003.18 ± 3.87 |
| 59 | 0.00 | 4.74 | 0.02 | 55.56 | 1993.55 ± 4.61 |
| 63 | 0.00 | 5.76 | 0.01 | 55.52 | 1992.68 ± 3.75 |
| 68 | 0.00 | 3.89 | 0.01 | 55.10 | 1983.57 ± 5.67 |
| 77 | 0.00 | 4.96 | 0.03 | 55.01 | 1981.67 ± 4.71 |
| 89 | 0.00 | 2.29 | 0.00 | 55.77 | 1998.08 ± 14.42 |
| 109 | 0.36 | 3.92 | 0.00 | 55.05 | 1982.37 ± 19.40 |
| 149 | 0.00 | 14.72 | 0.00 | 55.47 | 1991.53 ± 2.76 |
| 159 | 0.32 | 3.15 | 0.00 | 56.27 | 2008.85 ± 6.51 |
| 190 | 0.00 | 5.50 | 0.00 | 56.09 | 2005.04 ± 4.37 |
| Fuse | | 0.79 | 0.05 | 56.61 | 2016.20 ± 13.52 |
| Total age: | | | | | 1975.31 ± 1.32 |
| **GN amphibole** | | | | | |
| 91 | 42.8 | 0.24 | 0.00 | 190.26 | 3732.22 ± 33.23 |
| 220 | 6.33 | 0.54 | 0.00 | 80.81 | 2472.39 ± 25.71 |
| 350 | 3.03 | 1.84 | 0.23 | 81.71 | 2487.22 ± 8.30 |
| 370 | 1.47 | 0.78 | 0.82 | 59.91 | 2085.75 ± 16.08 |
| 405 | 0.84 | 1.09 | 0.00 | 57.86 | 2042.89 ± 14.45 |
| 450 | 0.74 | 2.05 | 1.08 | 57.27 | 2030.38 ± 7.41 |
| 500 | 0.00 | 3.41 | 2.4 | 56.85 | 2021.39 ± 4.91 |
| 542 | 0.04 | 14.23 | 3.80 | 56.61 | 2016.18 ± 2.86 |
| 557 | 0.03 | 13.57 | 3.82 | 56.81 | 2020.55 ± 2.86 |
| 567 | 0.36 | 5.80 | 4.07 | 56.18 | 2006.97 ± 4.18 |
| 593 | 0.17 | 6.80 | 4.01 | 56.81 | 2020.47 ± 3.22 |
| 636 | 0.27 | 15.79 | 4.39 | 57.07 | 2026.14 ± 2.72 |
| 669 | 1.74 | 1.62 | 3.31 | 55.87 | 2000.23 ± 10.66 |
| 748 | 0.49 | 6.42 | 4.12 | 56.78 | 2019.85 ± 3.73 |
| 548 | 0.62 | 2.59 | 4.32 | 55.92 | 2001.35 ± 8.14 |
| 1048 | 0.28 | 8.43 | 4.79 | 57.19 | 2028.55 ± 2.97 |
| Fuse | 0.19 | 14.8 | 4.43 | 56.94 | 2023.35 ± 2.59 |
| Total age: | | | | | 2040.52 ± 1.00 |

Table 2 (*Continued*)

| Laser step (mV) | Atmospheric (%) | $^{39}Ar_K$ (%) | $^{37}Ar_{Ca}/^{39}Ar_K$ (%) | $^{40}Ar^*/^{39}Ar_K$ (%) | Age (Ma) |
|---|---|---|---|---|---|
| **GN biotite** | | | | | |
| 25 | 79.76 | 0.14 | 0.00 | 36.33 | 1518.60 ± 91.34 |
| 38 | 57.77 | 0.12 | 0.00 | 33.56 | 1438.43 ± 81.93 |
| 83 | 30.54 | 0.90 | 0.00 | 34.92 | 1478.41 ± 17.17 |
| 103 | 15.67 | 2.05 | 0.01 | 52.41 | 1923.94 ± 6.29 |
| 113 | 2.56 | 4.41 | 0.00 | 54.30 | 196591 ± 3.42 |
| 118 | 1.20 | 4.28 | 0.00 | 54.54 | 1971.19 ± 3.22 |
| 123 | 1.15 | 3.30 | 0.00 | 54.75 | 1975.94 ± 3.57 |
| 139 | 0.80 | 4.14 | 0.01 | 54.74 | 1975.70 ± 4.21 |
| *155* | *0.41* | *4.18* | *0.01* | *53.12* | *1984.00 ± 0.00* |
| *170* | *0.45* | *4.72* | *0.01* | *54.99* | *1981.23 ± 3.19* |
| 185 | 0.43 | 3.33 | 0.00 | 54.48 | 1969.89 ± 4.36 |
| 215 | 0.25 | 6.02 | 0.00 | 54.74 | 1975.75 ± 2.85 |
| 238 | 0.44 | 5.38 | 0.01 | 54.54 | 1971.25 ± 3.25 |
| 253 | 0.18 | 4.06 | 0.00 | 54.67 | 1974.08 ± 3.28 |
| 270 | 0.30 | 3.91 | 0.00 | 54.83 | 1977.73 ± 3.60 |
| 290 | 0.13 | 5.18 | 0.00 | 54.64 | 1973.45 ± 2.67 |
| 309 | 0.27 | 3.83 | 0.00 | 54.53 | 1971.07 ± 4.43 |
| 335 | 0.12 | 10.69 | 0.00 | 54.55 | 1971.57 ± 3.25 |
| 343 | 0.00 | 3.80 | 0.00 | 54.82 | 1977.45 ± 3.40 |
| 356 | 0.22 | 2.98 | 0.00 | 54.68 | 1974.32 ± 4.35 |
| 378 | 0.31 | 3.69 | 0.00 | 54.48 | 1969.89 ± 3.10 |
| 418 | 0.38 | 4.71 | 0.00 | 54.56 | 1971.71 ± 3.37 |
| 495 | 0.05 | 6.28 | 0.00 | 54.63 | 1973.29 ± 2.85 |
| 591 | 0.35 | 4.96 | 0.00 | 54.59 | 1972.49 ± 3.50 |
| Fuse | 0.59 | 2.18 | 0.02 | 54.54 | 1971.38 ± 6.28 |
| Total age: | | | | | 1967.76 ± 0.79 |

of a perfectly linear superposition of induced magnetic moment by increasing and decreasing magnetic fields (Fig. 4e), indicating the strong dominance of the paramagnetic minerals in these rocks.

The above observations suggest that the magnetite is the principal ferromagnetic mineral in tonalite and meta-ultrabasite and the total absence of ferromagnetic minerals in Paramaca rocks, or very weak concentration of magnetite.

## 5. Paleomagnetic results

A pilot study was carried out on several specimens using both thermal and alternating magnetic field (AF) demagnetizations with a Pyrox furnace and an automated three-axis tumbler AF demagnetizer (LDA-3, AGICO geofysica), respectively. About 10 (AF, 1 to 100 mT) to 15 (thermal, ~20 to 595°C) progressive steps were applied to the demagnetization procedure.

### 5.1. Paramaca

Amphibolites and metasediments presented unimodal NRM intensities and susceptibilities ranging from 0.1 to 10 mA $m^{-1}$ and 2 to $7 \times 10^{-4}$ SI, respectively. Most of the Paramaca rocks were treated with the AF demagnetization method because of the unstable magnetic behavior during thermal treatment. A typical demagnetization response of these rocks is plotted in Fig. 5a. Initially, the magnetic remanence decays linearly until about 8 mT, then the direction becomes random and no characteristic component could be isolated from higher fields. Moreover, this low coercive magnetization could be observed only in five out of ten Paramaca sites, and directional analyses using the principal component analysis

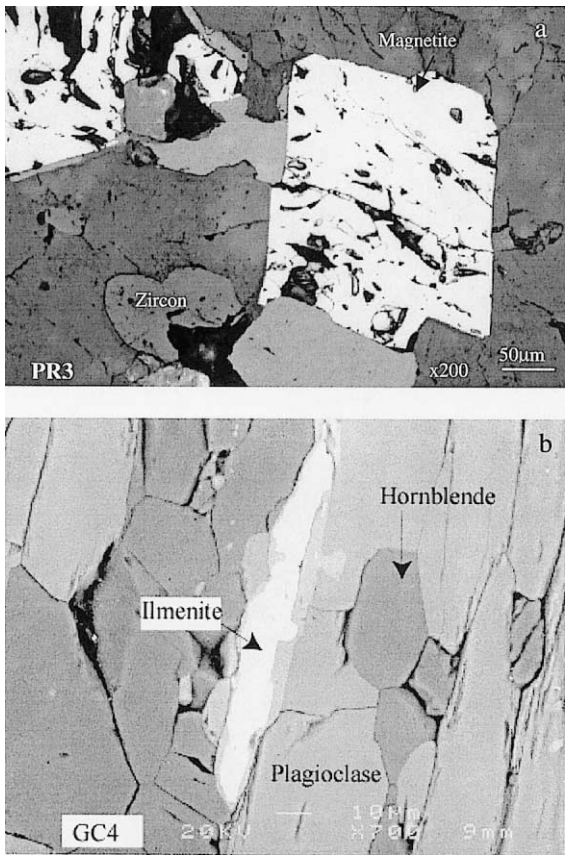



Fig. 3. (a) Reflected light photograph of a large subautomorphous titanomagnetite grain in tonalite from site GN. (b) Large-scale image in secondary electrons (SEM) of ilmenite from site GC4.

(Kirschvink, 1981) show dispersed directions within each site with poor statistical parameters (Table 1; Fisher, 1953). A mean direction was calculated from five sites: Dec = 349.8°, Inc = 23.6°, $k = 18.7$ and $\alpha_{95} = 18.2°$, which is not significantly different from the present Earth field ($D = 342.5°$, $I = 18.5°$, Fig. 6a). As described in the Section 4, no magnetite grains were found and the viscous behavior of these samples precludes any further discussion.

### 5.2. Tonalite and meta-ultrabasite rocks

NRM intensities vary from 100 to 800 mA $m^{-1}$ and susceptibilities range from $9 \times 10^{-4}$ to $2 \times 10^{-2}$ SI. They were thermally treated because complete demagnetization of NRM could be achieved. The representative results are plotted in orthogonal vector diagrams (Zijderveld, 1967) (Fig. 5b and c). For both tonalite and meta-ultrabasite rocks, after the removal of a low unblocking temperature random component at ~150°C (Fig. 5b), the direction becomes stable. The remanence is unblocked above 540°C and decays linearly to the origin until 580°C. The characteristic directions, namely OYA, are dominated by moderately steep downward inclinations with southeastward declination, except for site PS. The magnetic directions within each site are well grouped, and thus mean directions are computed by Fisher statistics (Fig. 6b, Table 1). The mean directions are distinct from the present Earth's field and the geocentric axial dipole field. The Fisher parameter $k$ is usually greater than 30 with $\alpha_{95}$ less than 10° (Table 1). These directions are also distinct from the local Early Jurassic dike paleomagnetic components characterized by a north-eastward declination and low inclination (Fig. 6b, Nomade et al., 2000). The above observations and the presence of subautomorphous magnetite grains with no evidence of a later destabilization (see Section 4) suggest that the magnetic remanence is of primary origin and represents a Paleoproterozoic magnetization. Site PS was sampled in the middle of the Oyapok River on a very limited outcrop. This outcrop was perhaps not in situ and could be a free block. Therefore, this site was excluded from the mean direction calculation: Dec = 133.8°, Inc = 60.2°, $k = 59.9$, $\alpha_{95} = 9.9°$ and $N = 5$ (Fig. 6b). The corresponding virtual geomagnetic pole (VGP) is OYA: $\lambda = 28.0°$S, $\phi = 346.0°$E, $k = 31.9$, $A_{95} = 13.8°$ and $N = 5$ (Table 1).

## 6. $^{40}Ar/^{39}Ar$ results

Analytical data for four dated specimens are given in Table 2. Fig. 7 shows age and $^{37}Ar_{Ca}/^{39}Ar_{K}$ ratio spectra for minerals from B107 (Fig. 7a and b) and site GN (Fig. 7c and d).

Amphibole single grains (green grains) yield a very flat spectrum for the B107 and GN speci-

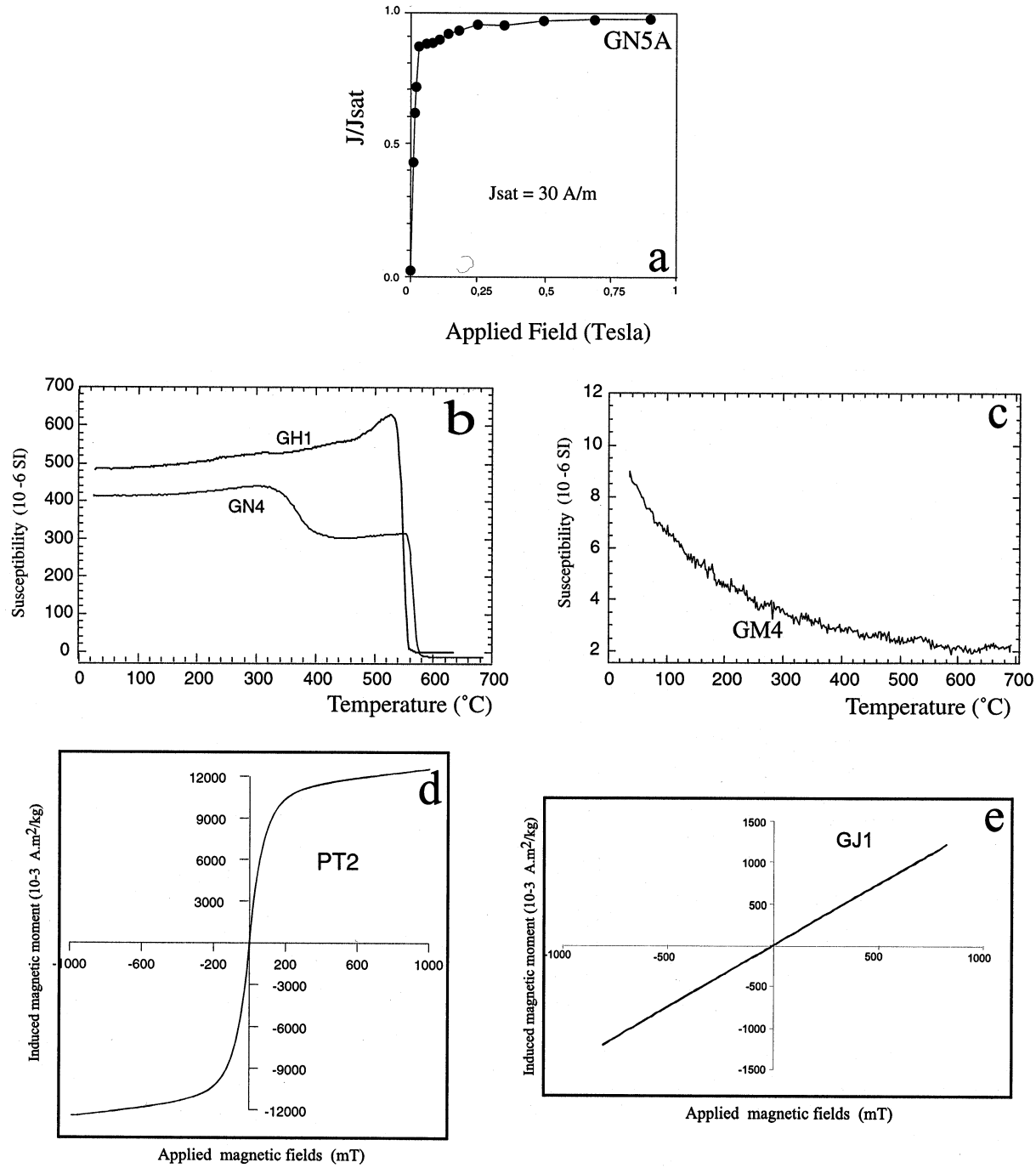



Fig. 4. Typical isothermal remanent magnetization (IRM) acquisition for tonalite and meta-ultrabasite rocks (a). Typical curves of the thermomagnetic experiments for the tonalite, meta-ultrabasite (b) and Paramaca rocks (c). Hysteresis curves performed on the tonalite rocks showing a narrow-waisted and typical of low coercivity magnetite grains (d) and (e) Paramaca rocks showing a perfectly linear superposition of the two induced magnetic moment curves produced by increasing and decreasing magnetic fields.

mens (Fig. 7a and b), after a sharp decrease in age at low temperature. The corresponding $^{37}Ar_{Ca}/^{39}Ar_K$ ratios remain constant throughout the flat section of spectra (Fig. 7a and b) and indicate that pure amphiboles were analyzed. The plateau age is $2052 \pm 4$ Ma (88% $^{39}$Ar released) for B107. The GN specimen does not show a plateau age following our criteria defined above. The flatness of the spectrum on more than 90% of $^{39}$Ar leads us to interpret the corresponding weighted mean age with 93% $^{39}$Ar released of $2020 \pm 4$ Ma as reliable.

Biotite single grains display age spectra (Fig. 7c and d) characterized by (1) a sharp increase of age at low temperature followed by a more or less flat section over at least 80% of $^{39}$Ar released. In the case of B107, this section is characterized by a slight increase of ages (step 6–7) followed by a decrease. This is typical of slightly chloritized biotites affected by $^{39}$Ar recoil during irradiation (Ruffet et al., 1991). It is therefore likely that the weighted mean of $1995 \pm 4$ Ma, calculated on 80.3% $^{39}$Ar, is reliable. For the GN biotite age, the very flat section over 87% of $^{39}$Ar (19 steps) is only altered by two steps (9 and 10) that are clearly the results of slight $^{39}$Ar recoil. Therefore, we calculated a plateau age of $1973 \pm 4$ Ma with 17 steps excluding the discordant steps 9 and 10.

## 7. Discussion

### 7.1. Age of the magnetic remanence

The significant difference of 47 Ma in $^{40}Ar/^{39}Ar$ ages observed in amphiboles and biotites for site GN in the central Oyapok zone shows that the GN tonalite rock cooled down from about 500–550°C to about 250–350°C at about $2020 \pm 4$ Ma and $1973 \pm 4$ Ma, respectively, according to the proposed biotite and amphibole K/Ar blocking temperatures (Harrison, 1981; Harrison et al., 1985). It is important to notice that the compositional effect in hornblende K/Ar closure temperature (Dahl, 1996) suggests a temperature of $500 \pm 5$°C for Mg-rich amphibole. Nevertheless, this high precision isotopic closure temperature is not adequate because of the many uncertain factors that can be induced by K diffusion in amphibole. For the biotite the compositional effect in closure temperature is well documented by Grove and Harrison (1996), but this effect is unpredicted and does not allow the constraint of the uncertainties in the isotopic closure temperature. For these reasons, we have used the blocking temperatures proposed above (Harrison, 1981; Harrison et al., 1985).

A similarly important difference in ages (57 Ma) between these two minerals is also dis-

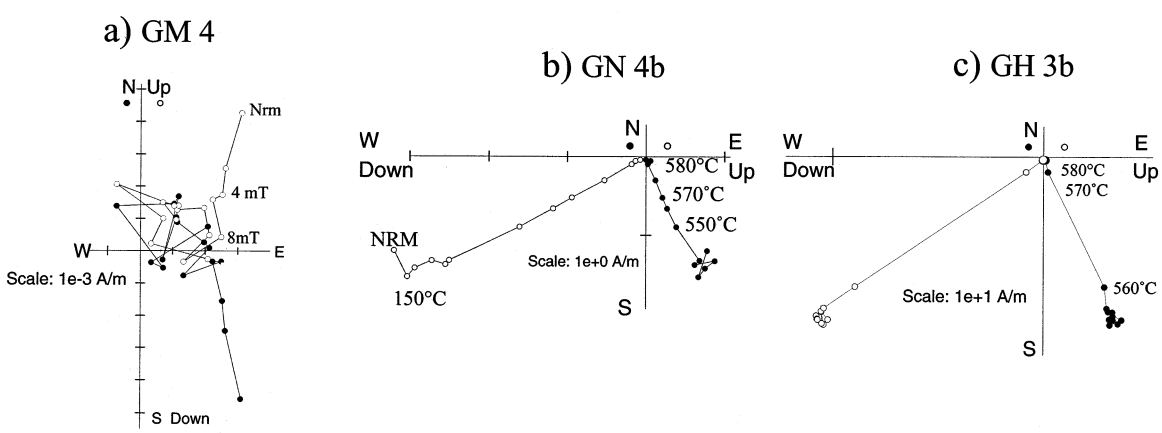



Fig. 5. Representative orthogonal vector diagrams (Zijderveld, 1967) of progressive thermal and alternating field demagnetizations. The closed (open) symbol refers to the horizontal (vertical) plane. (a) Low coercive remanent direction in Paramaca. (b) and (c) Typical SE declinations with medium to high positive inclinations for tonalite and meta-ultrabasite rocks.

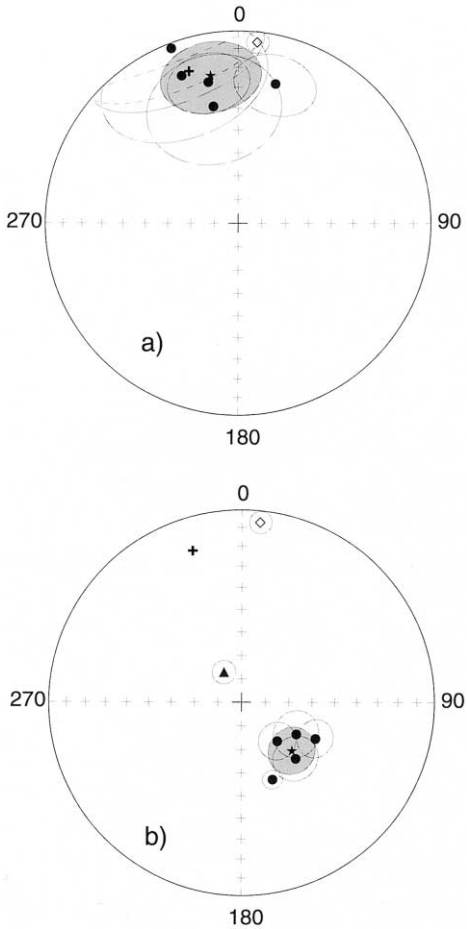



Fig. 6. Equal-area stereoplot of mean direction: (a) Paramaca; (b) tonalite and meta-ultrabasite. The star stands for the mean direction with corresponding 95% confidence oval, cross for the present Earth field and diamond shape for local Early Jurassic pole (Nomade et al., 2000). The triangle represent the mean direction of site PS (excluded from the average).

played in a separate pluton (B107 specimen; $2052 \pm 4$ Ma and $1995 \pm 4$ Ma, for amphiboles and biotites respectively), suggesting that low cooling rate was homogeneous during the Paleoproterozoic in this zone.

The magnetic remanence was acquired by the magnetite grains during the protracted cooling of the intrusion. The age of magnetization must, therefore, lie between the crystallization age of the rocks and the ($^{40}Ar/^{39}Ar$) biotite age $1973 \pm 4$ Ma in site GN. $^{40}Ar/^{39}Ar$ data are plotted on a $T/t$ diagram (Fig. 8) in order to estimate tentatively the magnetic remanence age for the pole OYA. A relatively slow cooling rate of $4.8 + 2.6/-2.1$°C Ma$^{-1}$ is deduced from these ages. Despite the possible bias between K/Ar and Pb/Pb ages due to a poor precision on the $^{40}K$ decay constants (Min et al., 2000), we plotted the Pb/Pb ages of the French Guyana plutonic activity (Pb/Pb ages, Vanderhaeghe et al., 1998; Fig. 8). This rate is in agreement with a low unpredictable rate starting between 2130 and 2080 Ma ($< 800$°C, Lee et al., 1997; Dahl, 1997). This cooling rate is in agreement with that from greenstone-belt granites of similar age in the São-Fransisco craton (Chauvet et al., 1997) and can be interpreted as the result of gradual exhumation. By using magnetic blocking temperatures between 540 and 580°C, which corresponded to the unblocked temperatures of our samples (see above), we obtain for site GN a magnetic remanence age of $2036 \pm 14$ Ma for the pole OYA.

### 7.2. Paleogeographic and geodynamic implications

The new paleomagnetic pole (OYA) from French Guyana and available Paleoproterozoic paleomagnetic data from the nearby zone (Venezuela) of the Guyana Shield are listed in Table 3. The results from Imataca granulite (Im 1 and Im 2; squares in Fig. 9a) and Encrucijada granite (En A1 and En A2; circles in Fig. 9a) show consistent poles and Rb/Sr ages, with due consideration of statistical errors (Fig. 9a). This suggests that the GFZ has experienced no significant activity since 2000 Ma from the point of view of paleomagnetism. Therefore, a mean pole for this area was calculated (Table 3, Fig. 9a). Fig. 9a shows that pole OYA is statistically distinct from the other four poles from Venezuela, and from their mean poles with an angular difference of $32.6 \pm 21.5$°. The difference between these two poles reflects a paleolatitudinal difference ($31.8 \pm 21.5$°) with consistent declinations ($0.6 \pm 23.9$°). Two hypotheses may be proposed to explain this difference: (i) if the ages of magnetic remanence are similar among the paleomagnetic data, over 1000 km of latitudinal crustal shortening had occurred between the Venezuela and French

Table 3
Available Paleoproterozoic poles for the Guyana Shield and their possible paleomagnetic corresponding ages. Plat. and Plong.: latitude and longitude of polar coordinates; $N$, $\alpha_{95}$, Lat.: number of site, confidence interval at 95% level, paleolatitude

| Rock | | Plat. | Plong. | $\alpha_{95}$ | $N$ | Lat. | Methode of dating | Magnetie remanence age (Ma) | Reference |
|---|---|---|---|---|---|---|---|---|---|
| OYA | (tonalite, meta-ultrabasite) | −28.0° | 346.0° | 13.8° | 5 | 41°S | $^{40}Ar/^{39}Ar$ Amph: $2020 \pm 4$ Ma; $^{40}Ar/^{39}Ar$ Biot: $1973 \pm 4$ Ma | $2036 \pm 14$ | This study |
| Im 1 | granulite | −49.0° | 18.0° | 18.0° | 3 | 10°S | Rb/Sr isochron:$2020 \pm 60$ Ma (Montgomery and Hurley, 1978) | | Onstott and Hargraves (1981) |
| Im 2 | granulite | −29.0° | 21.0° | 18.0° | 3 | 13°S | | | |
| En A1 | granite | −55.0° | 8.0° | 6.0° | 5 | 1°S | Rb/Sr isochron: $2064 \pm 87$ Ma; $^{40}Ar/^{39}Ar$ Amph: $1972 \pm 4$ Ma; $^{40}Ar/^{39}Ar$ Biot: $1882 \pm 2$ Ma | | |
| En A2 | granite | −37.0° | 36.0° | 18.0° | 3 | 14°S | | $2000 \pm 10$ | Onstott et al. (1984) |
| Mean of Venezuela poles | | 43.0° | 22.0° | 16.5° | 4 | 9°S | | | |

Guyana blocks separated by the PJF since that age (relative movement); (ii) if the magnetic remanence ages of these two blocks are different, the same order of latitudinal displacement of the blocks occurred during the corresponding period (absolute movement).

The Rb/Sr dating method was applied to both Imataca granulite and Encrucijada granite and yielded concordant ages of $2020 \pm 60$ Ma and $2062 \pm 87$ Ma (Onstott and Hargraves, 1981; Onstott et al., 1984, 1989). The new age of pole OYA ($2036 \pm 14$ Ma) is included within the statistical estimates of the previous two, but the statistical precision is too poor ($> 60$ Ma) to determine whether these ages are really consistent. However, the Encrucijada granite was also dated by the $^{40}Ar/^{39}Ar$ method on biotite and amphibole, which show two ages of $1975 \pm 4$ Ma and $1882 \pm 2$ Ma ($1\sigma$) respectively. Onstott et al. (1984) demonstrated that the isolated magnetic components from Encrucijada granite were carried by magnetite grains. In order to estimate the magnetization age for Encrucijada granite, we used similar isotopic and magnetic temperature closures

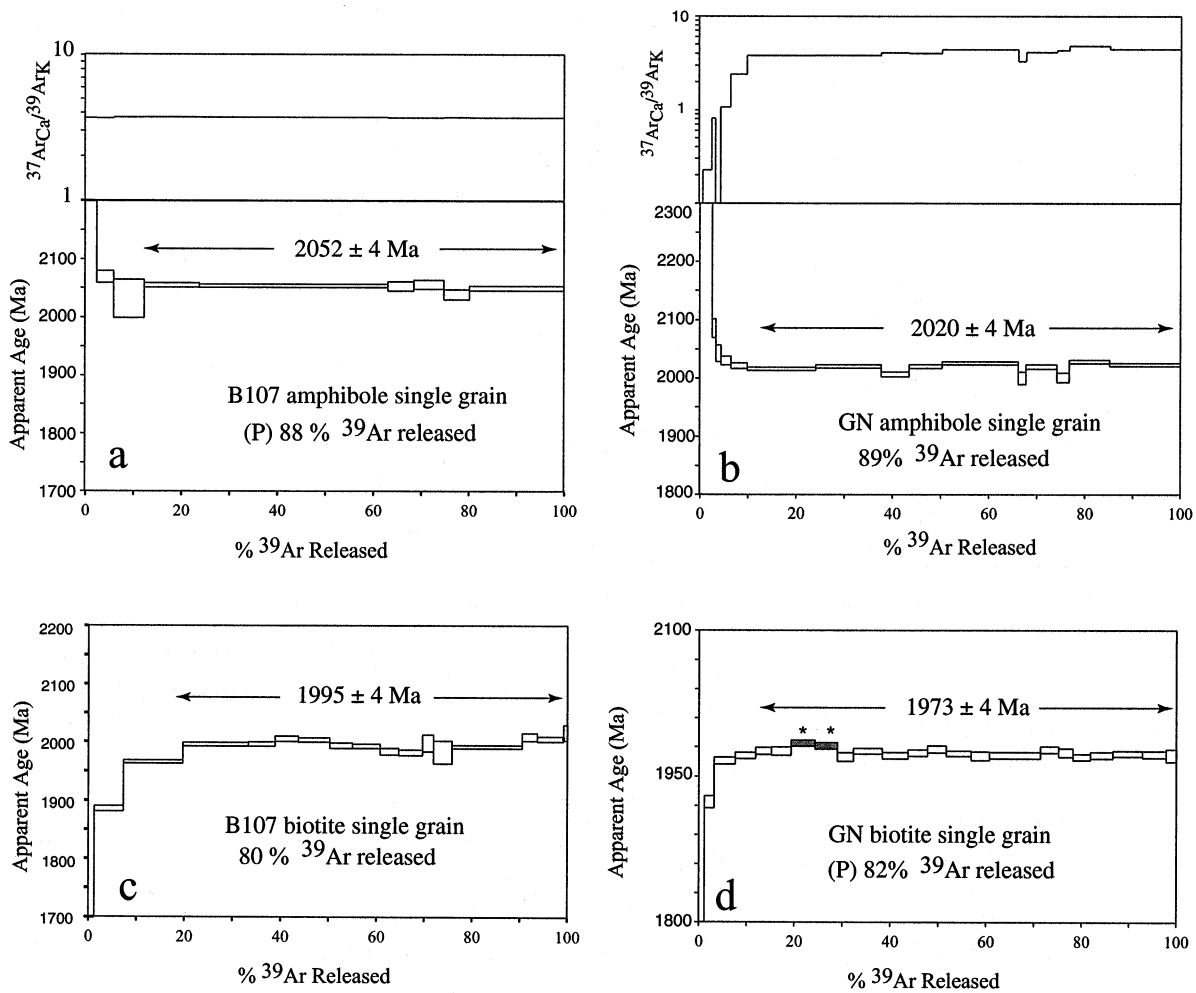



Fig. 7. $^{40}Ar/^{39}Ar$ age spectra for amphibole and biotite from tonalite (a and c for B107 sample; b and d for the tonalite of the paleomagnetic site GN). $^{40}Ar_{Ca}/^{39}Ar_K$ ratio spectra displayed by amphiboles are also shown. P indicates plateau ages. * Indicates the steps probably affected by $^{39}Ar$ recoil due to slight chloritization of biotite, and excluded from the plateau age calculation.

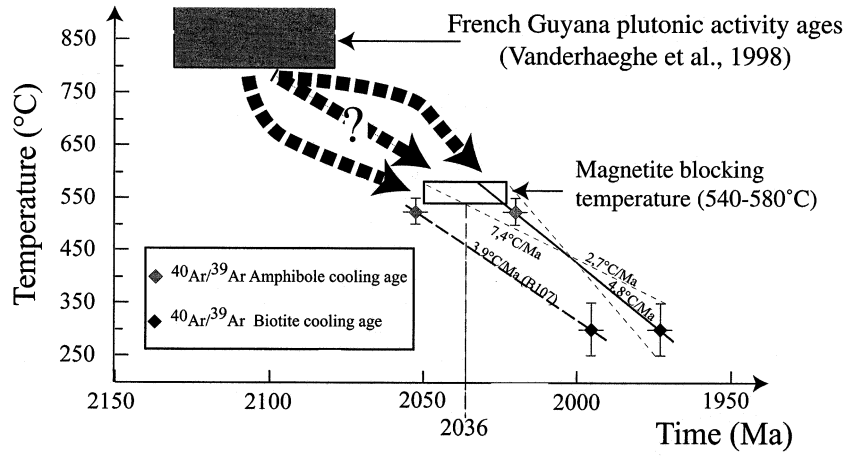



Fig. 8. Temperature–time diagram showing the inferred cooling paths for the two sampling zones: B107 in dotted line and GN in solid line. Previous Pb/Pb geochronological investigations for French Guyana (Vanderhaeghe et al., 1998) are also indicated.

for the amphibole, biotite and magnetite as described above. The magnetization age for the Encrucijada pluton was found to be $2000 \pm 10$ Ma (Table 3) with an estimated cooling rate of ca $2 \pm 1$°C Ma$^{-1}$ (Swapp and Onstott, 1989). This age appears significantly different from that of pole OYA (about $36 \pm 16$ Ma). The difference of about 1.6% in age between these two groups of paleomagnetic data is significant because the two series of minerals were irradiated with the same flux monitor Hb3gr, which is appropriate for this age domain, and in the same nuclear reactor.

Geographically, the two paleomagnetic sampling zones (Venezuela and French Guyana) are separated by the senestral PFZ (Figs. 1 and 9b) and the CGGB. The age of granulitization for the core of the Falawatra complex part in the CGGB in Suriname was estimated between 2006 to 2046 Ma (U/Pb, Bosma et al., 1983). According to these paleomagnetic data the paleogeographical reconstruction indicates that these fault zones had a nearly east–west orientation at about 2.0 Ga (Fig. 9b). Therefore, the possible longitudinal displacement along this mobile zone between the two blocks at that time cannot be detected by paleomagnetism. Moreover, no high-pressure metamorphism ($< 8$ kbar) and/or important shortening are known in the CGGB zone separating the two blocks (Kroonenberg, 1976).

Based on geological and geochronological data, the $31.8 \pm 21.5$° latitudinal difference between the French Guyana block at $2036 \pm 14$ Ma and Venezuela block at $2000 \pm 10$ Ma may correspond to a northward movement (in respect of the pole polarity proposed by Onstott et al. (1984)) of the Guyana Shield from 40°S to 10°S. The rate of this northward displacement is estimated as $9 \pm 7$ cm/year, which is similar to present high-velocity plate movement.

## 8. Conclusion

The paleomagnetic and geochronological data from this and previous studies lead to the following conclusions.

(1) Subautomorphous magnetite is the principal magnetic remanence carrier for tonalite and meta-ultrabasite from our sample collection. In the majority of the greenstones (Paramaca formation) the ferromagnetic minerals are rare or absent.

(2) Two magnetic components were obtained, one from the Paramaca rock (P) and one from tonalite, meta-ultrabasite rocks (OYA). Unstable Paramaca magnetic components reflect a viscous overprint of recent origin. We found that component OYA is of primary origin and represents a Paleoproterozoic magnetization. Pole OYA was

therefore computed from OYA components: $\lambda = 28.0°S$, $\phi = 346.0°E$, $k = 31.9$ and $A_{95} = 13.8°$ and $N = 5$.

(3) $^{40}Ar/^{39}Ar$ results show systematic and similar differences between amphibole and biotite ages on two dated rocks; the slow cooling rate of ca $4.8 + 2.6/-2.1°C\ Ma^{-1}$ of the rocks was interpreted between about 525 and 300°C as being due to the gradual exhumation of the Oyapok–Camopi river zone. A magnetization age of $2036 \pm 14$ Ma from tonalite and meta-ultrabasite was extrapolated from this slow cooling rate.

(4) The new paleomagnetic pole OYA from French Guyana is significantly distinct in latitude

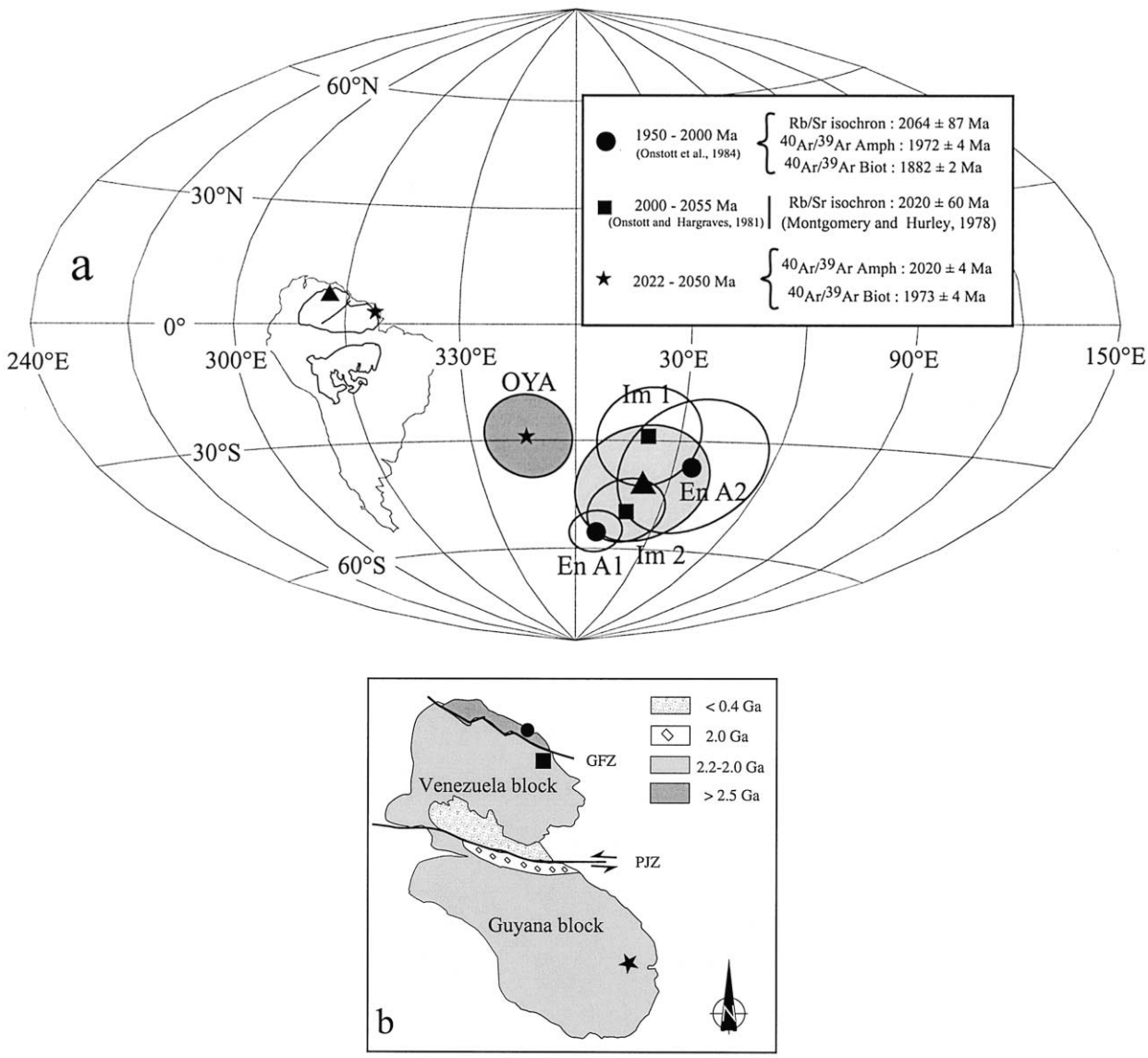



Fig. 9. (a) Aitoff Hammer projection of the paleomagnetic poles from the present study (star) and Venezuela (triangle) (Onstott and Hargraves, 1981 (square); Onstott et al., 1984 (circle)). (b) The Paleoproterozoic orientation of the Guyana Shield with respect to the paleo-north.

from that of Venezuela (Encrucijada pluton), with an estimated magnetization age of $2000 \pm 10$ Ma. This difference of 1.6% in age between the two blocks is probably significant and may be the result of a northward movement of the Guyana Shield of $31.8 \pm 21.5°$ from about 2036 to 2000 Ma with an estimated rate of $9 \pm 7$ cm/year.

(5) Relative movement between the French Guyana and Venezuela blocks along the Pisco Jurua fault zone cannot be estimated during this period by paleomagnetic investigations owing to the east–west orientation of this mobile zone during the 2036–2000 Ma period.

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